

Transcutaneous electrical nerve stimulation on acupoints reduces fentanyl requirement for postoperative pain relief after total hip arthroplasty in elderly patients

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ABSTRACT

Background. Transcutaneous electrical nerve stimulation (TENS) is regarded as an effective treatment for various types of pain. However, no randomized controlled trial has investigated TENS on acupoints for postoperative analgesia in elderly patients. This study aim to investigate whether TENS on acupoints has any favorable effect on complementary analgesia after total hip arthroplasty (THA) for elderly patients compared with a sham control treatment.

Methods. Sixty-eight elderly patients requiring THA surgery were enrolled and randomly allocated to one of two groups. Group Acu received true TENS on acupoints (bilateral P6, L14; ST36, GB31 ipsilateral to the surgery site) and Group Sham received sham treatment. All patients received patient-controlled analgesia for two days postoperatively. Analgesia was assessed by postoperative fentanyl requirement and pain intensity using a visual analogue scale (VAS-10 cm). The incidence of analgesia-related side effects, optional medication use and effects of patients' blinding were recorded.

Results. Fentanyl consumption in Group Acu was lower than that in Group Sham at 24 h (mean $\pm$ SD; 360 $\pm$ 117 vs. 572 $\pm$ 132 μ g; $P < 0.001$) and 48 h (712 $\pm$ 184 vs. 1022 $\pm$ 197 μ g; $P < 0.001$) after surgery. Postoperative pain intensity measured by VAS was similar in both groups. The incidence of opioid-related side effects and rescue medication for postoperative analgesia was significantly higher in Group Sham than in Group Acu. Differences between the groups regarding the effects of patients' blinding were not significant.

Conclusion. TENS on specific acupoints is an effective and complementary approach to reduce postoperative analgesic requirement in elderly patients after THA. (*Minerva Anesthesiol* 2012;78:887-95)

Key words: Pain, postoperative - Analgesia - Arthroplasty, replacement, hip - Elderly.

Acute postoperative pain is a challenge to clinicians for postoperative care, especially in elderly patients with comorbidities such as coronary heart disease. Elderly patients undergoing total hip arthroplasty (THA) frequently experience high levels of postoperative pain.^{1, 2} If postoperative pain is not managed effectively in these patients, postoperative complications

such as myocardial ischemia, pulmonary atelectasis and postoperative cognitive dysfunction (POCD)³⁻⁶ may develop as pain and its associated pathological changes will be harmful for our body and organs to be functional normally. The use of patient-controlled intravenous analgesia (PCA) is considered to be safe and efficient for elderly patients.⁷ However, PCA could not only

lead to numerous opioid-related side effects such as respiratory depression, nausea, vomiting, pruritus and dizziness, but also may increase a risk of the POCD,⁸ which are potential risk factors of postoperative morbidity and even mortality.⁹ Therefore, it is important to explore alternative postoperative pain relief strategies for elderly patients with fewer side effects than those currently available.

Transcutaneous electrical nerve stimulation (TENS) has been reported to significantly decrease opioid requirements within 24 h postoperatively,¹⁰ although it can be ineffective in the treatment of postoperative acute pain in certain cases, including total knee arthroplasty.¹¹ Moreover, TENS treatment on acupoints has better pain relief efficacy and can reduce the incidence of postoperative nausea and vomiting considerably.^{10, 12}

The aim of the present study was to investigate whether TENS on specific acupoints can provide complementary analgesia to reduce the requirement for fentanyl PCA and its related side effects after THA in elderly patients.

Materials and methods

Study design and randomization

After obtaining Ethics Committee approval of Xuanwu Hospital, Capital Medical University, Beijing, China, 68 elderly patients scheduled for THA with combined spinal epidural anesthesia (CSEA) were recruited for the study. The study was performed at the Departments of Anesthesiology and Orthopedics, Xuanwu Hospital, Capital Medical University, China, from June 2009 to March 2011. Preoperative anesthesiologic evaluation and the patients' randomization for the study were performed the day before surgery. After obtaining written informed consent, all patients were divided into two groups by use of a computer-generated random number table: Group Acu-PCIA plus TENS therapy for 30 min before incision, and at 2 h, 4 h, 20 h and 44 h after surgery; Group Sham (sham control) - PCIA plus sham TENS therapy (all conditions are identical but without electrical stimulation).

Patient selection

Patients who are over 65 years old, with an ASA Grade I-III scheduled for elective THA for hip fracture were recruited in the study. The patients were excluded if they: 1) had a history of acupuncture treatment; 2) were not capable of understanding how to use the PCA pump, visual analogue scale (VAS) for pain measurement or consent form; 3) had a history of alcohol abuse; 4) had a history of opioid medication; 5) had psychiatric illness; 6) suffered from extreme obesity (Body Mass Index >40 kg/m²); 7) had local or systemic infection; 8) or if they had cardiac pacemakers.

TENS, sham TENS and specific acupoints

Practicing acupuncturists of at least two years' experience, marked the specific points of the patients in both groups the evening before surgery (Figure 1). In both groups, six acupoints were selected: bilateral P6, L14 (P6: on the anterior surface of the forearm, 2-finger widths up from the first wrist crease and between the tendons of flexor carpi radialis and palmaris longus; L14: on the dorsum of hand, between the 1st and 2nd metacarpal bones, in the midpoint of 2nd metacarpal bone on the radial side; ST36: one finger-breadth from the anterior crest of the tibia; GB31: at the junction of the lateral 1/3 and medial 2/3 of the distance between the greater trochanter of the femur and the hiatus of the sacrum).

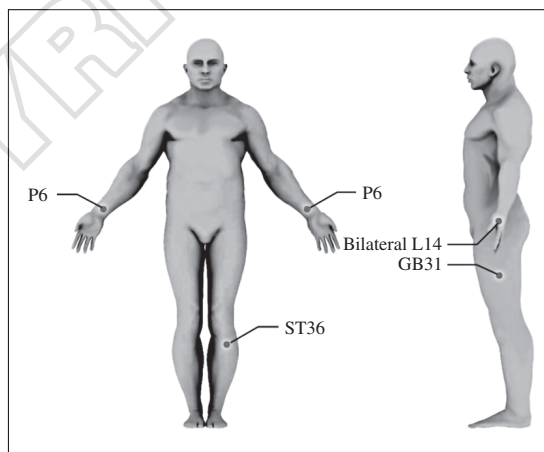


Figure 1.—Anatomical sites of acupuncture stimulated in the study. In both groups, six acupoints were selected: bilateral P6, L14 and ipsilateral to the surgery site ST36, GB31. (P6: on the anterior surface of the forearm, 2-finger widths up from the first wrist crease and between the tendons of flexor carpi radialis and palmaris longus; L14: on the dorsum of hand, between the 1st and 2nd metacarpal bones, in the midpoint of 2nd metacarpal bone on the radial side; ST36: one finger-breadth from the anterior crest of the tibia; GB31: at the junction of the lateral 1/3 and medial 2/3 of the distance between the greater trochanter of the femur and the hiatus of the sacrum).

dons of flexor carpi radialis and palmaris longus; L14: on the dorsum of hand, between the 1st and 2nd metacarpal bones, in the midpoint of 2nd metacarpal bone on the radial side) and ipsilateral to the surgery site ST36, GB31 (ST36: one finger-breadth from the anterior crest of the tibia; GB31: at the junction of the lateral 1/3 and medial 2/3 of the distance between the greater trochanter of the femur and the hiatus of the sacrum). The choice of the specific acupoints was based on previous clinical studies.^{10, 13, 14} Six cutaneous self-adhesive electrode pads (HANS electrodes; Gesun) sized 16 cm² were attached on the six acupoints marked by the acupuncturists. All the electrodes were connected to the HANS device (HANS-200B six-channel unit; Gesun Medical Technology CO. LTD, Nanjing, PRC) and stimulated in the standard dense-and-disperse (D-D) mode at a frequency setting of 2/100 Hz (*i.e.*, alternately stimulating at 2 Hz and 100 Hz every 3 seconds). The intensity of stimulation was set at 0 mA for group Sham and at 9-20 mA for group Acu (depending on the patient's ability to tolerate the stimulation). All patients received the stimulation 30 min before incision and at several time points in the 2 days after surgery: at 2 h, 4 h, 20 h and 44 h. Patients in both groups were told that they may or may not experience the electrical stimulation, however, the "in use" light on the HANS stimulator was flashing as usual in group Sham when it was activated. TENS treatment on these acupoints continued for two days after surgery because the usual period of maximal postoperative pain after THA was about two days. The duration of patient-controlled analgesia *via* PCA pump for all patients after elective THA in our hospital was also two days. Only acupuncturists who performed TENS were aware of patients' allocation, however, interaction between the patient and acupuncturist was limited to the time required for attaching electrode pads on acupoints.

Combined spinal epidural anesthesia, postoperative care and PCA

Both groups of patients received combined spinal epidural anesthesia (CSEA) without pre-

medication. Electrocardiogram (ECG), pulse oxygen saturation (SpO₂), invasive blood pressure and temperature were monitored before anesthesia. The patients were injected with 1.5-2 mL of 0.5% Bupivacaine Hydrochloride into the subarachnoid space through the L3/4 intervertebral space, and then an epidural catheter was placed in the epidural space and advanced 3 cm. Combined spinal epidural anaesthesia was maintained below T-10 for all patients. All patients were not administered 2% lidocaine hydrochloride through epidural catheter unless spinal anesthesia block did not reach to surgical requirement level. In order to maintain stable anesthesia, heart rate (HR) and mean arterial pressure of all patients were kept within 20% of baseline values with fluid loading and administration of vasoactive drugs when necessary. An electric heating carpet was placed on the operating table to maintain body temperature. After surgery the epidural catheters were removed, and patients were transferred to the orthopedic ward and ECG, HR and SpO₂ were monitored for 24 hours.

At the end of operation, the PCA pump (Smith Medical MD, Inc - Model 6300-CADD-Legacy Ambulatory Infusion Pump) with intravenous fentanyl (10 µg/mL) was connected to the patient for two days and 2 mL fentanyl was infused as the initial bolus. The setting of the PCA pump was: background infusion 2 mL/h, locking time 15 min, PCA 2 mL.

The aim of pain relief after surgery was a pain intensity of less than 4 on a 10 visual analogue scale (VAS-10cm, where 0=no pain; 10=worst pain imaginable). The patients were instructed how to use the PCA pump and encouraged to press the button every time they experienced pain >4 in order to inject the opioid analgesic via the pump. If patients still reported intensity of pain >4 on the first postoperative day, rescue medication (intramuscular pethidine hydrochloride 50 mg) was administered by the orthopedic surgeon and registered on the study protocol. The Orthopedic Surgery Department nurses and the physicians involved in patient management were blinded to the group allocation and had no previous TENS and acupuncture expertise.

Outcome measures

The primary outcome of this study was fentanyl consumption *via* PCA pump at 24 h after surgery, but total fentanyl requirement *via* PCA pump within 48 h was also recorded. At the same time, the data collected from patients included: 1) pain intensity on VAS-10 cm at rest and on movement (turning over to the healthy side in the bed) at 8 h, 24 h and 48 h postoperatively when they were awake before administration of rescue medication and two hours later after stimulation; 2) Ramsay sedation scale (RSS 1-6, where 1=anxious and agitated or restless, or both; 2=cooperative, oriented and tranquil; 3=responding to commands only; 4=exhibiting brisk response to light glabellar tap or loud auditory stimulus; 5=exhibiting a sluggish response to light glabellar tap or loud auditory stimulus; 6=exhibiting no response) at 24 h, 48 h after surgery; 3) the incidence of analgesia-related side effects: nausea and vomiting, excessive sedation (RSS $\geq$ 5), dizziness and pruritus; and 4) the frequency of rescue medication for pain relief during the two days after surgery. Special intraoperative situations such as hypotension, massive hemorrhage and hypothermia, duration of the surgery and the dosage of epidural local anesthetics were recorded. All data were collected by the investigator who has no previous knowledge concerning TENS and was blinded to the patients' group allocation.

Statistical analysis

Intravenous morphine can be converted to equivalent intravenous fentanyl by the conversion table.¹⁵ Therefore the sample size calculated for this study was based on published data indicating morphine consumed *via* PCA (mean $\pm$ SD=39.1 $\pm$ 17.9 mg) within 24 h in the placebo group in the similar population and surgical setting.¹⁶⁻¹⁸ It was estimated that a sample size of 27 patients per group would be sufficient to detect a difference of 35% between PCA plus TENS and PCA plus sham TENS in the total morphine consumption within 24 h postoperatively for a two-tailed $\alpha=0.05$ with a study power of 0.8. To account for an anticipated 20% drop-

out rate, we have managed to recruited sixty-eight patients in this study.

Statistical analysis was performed using SPSS 16.0 (Chicago, IL, USA). The Kolmogorov-Smirnov test was used to check whether data were normally distributed. The normally distributed continuous data (demographics, fentanyl requirement, pain intensity on VAS-10 cm, heart rate and blood pressure) were compared using the unpaired Student's *t*-test. Chi-square test was used to analyze the success of patients' blinding and the incidence of analgesia-related side effects. A *P* value <0.05 was considered a statistical significance.

Results

Patient characteristics

Sixty-eight patients who was recruited into in this study met the inclusion criteria and were randomized. Thirty patients per group had completed this study and their data were used for final data analysis. Their demographic data, duration of the surgery and dose of intraoperative epidural local anesthetic was summarized and there was no difference between the two groups (Table I). Eight patients were withdrawn during the study because of: 1) intraoperative complications leading to general anesthesia (severe bleeding requiring blood transfusion of more than 4 units of washed red blood cells and severe cardiovascular instability); 2) postoperative cognitive dysfunction.

TABLE I.—Patients' demographic data, duration of anesthesia and epidural dosage.

	Group Acu (N.=30)	Group Sham (N.=30)
Age (y)	76 $\pm$ 6	75 $\pm$ 5
Weight (kg)	66 $\pm$ 12	63 $\pm$ 14
Height (cm)	162 $\pm$ 8	163 $\pm$ 7
Gender (M/F)	14/16	13/17
Withdrawn (M/F)	1/2	2/3
Duration of anesthesia (min)	122 $\pm$ 30	124 $\pm$ 22
Dosage of epidural	170 $\pm$ 25.8	173 $\pm$ 11.5
Local anesthetics (mg)		

There was no difference between two groups in patients' demographic, duration of anesthesia and dosage of epidural lidocaine hydrochloride. Data are mean $\pm$ SD or number of patients. Group Acu: PCA+TENS on acupoints; Group Sham: PCA+Sham TENS on acupoints.

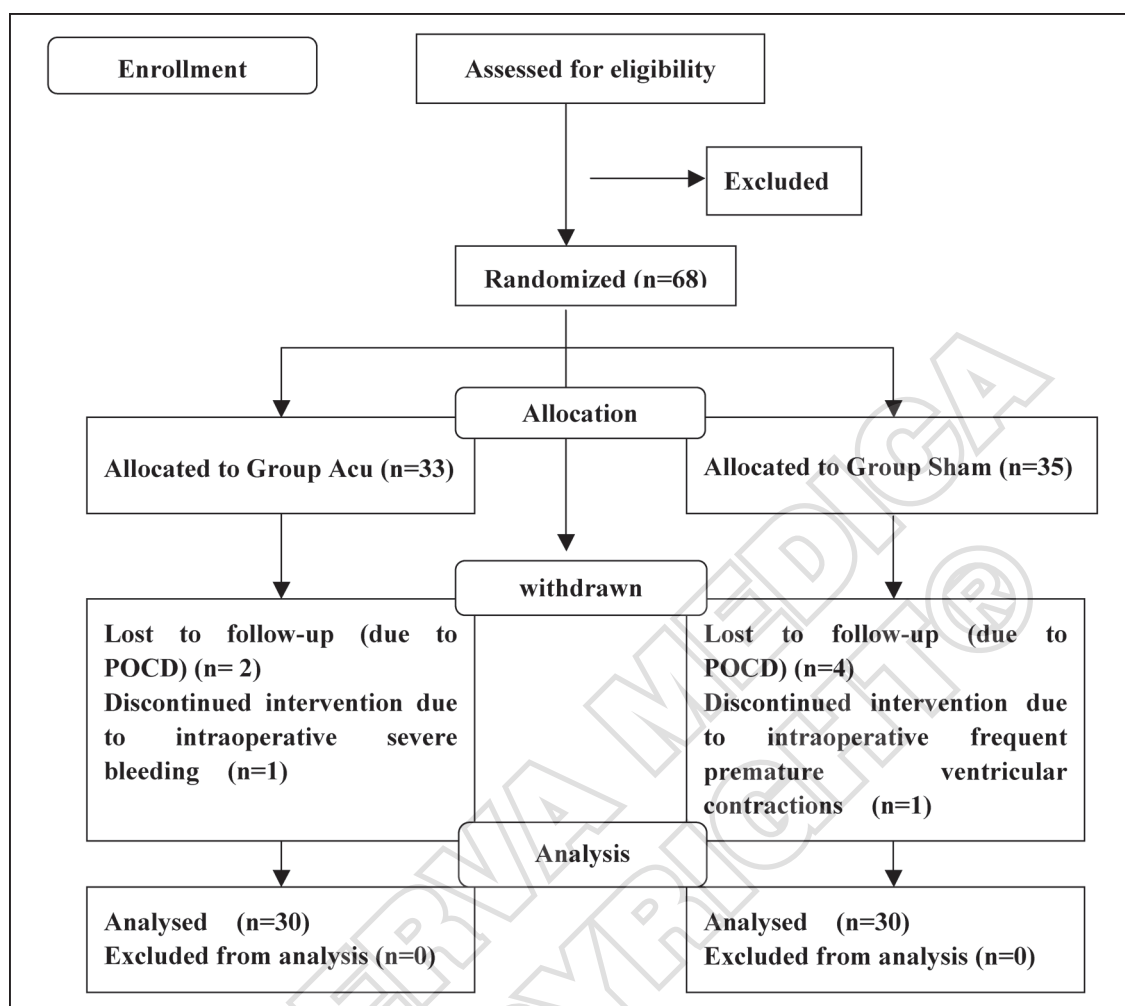


Figure 2.—Patients participated in this study. Sixty-eight patients were recruited and randomized into two groups. During the study, two patients in Group Acu and four patients in Group Sham lost to follow-up due to POCD. One patient in Group Acu discontinued intervention because of intraoperative severe bleeding, and one patient in Group Sham cease intervention due to intraoperative frequent premature ventricular contractions. Ultimately, 30 patients in each group completed this study. POCD: postoperative cognitive dysfunction. Group Acu: PCA+TENS on acupoints; Group Sham: PCA+Sham TENS on acupoints.

tion (Figure 2). The difference between the two groups regarding withdrawal rate was not statistically significant. Heart rate, respiratory rate and blood pressure were similar in both groups during surgery and the postoperative monitoring period (data not shown).

Analgesic consumption and pain intensity

The Kolmogorov-Smirnov test confirmed that the consumption of fentanyl by PCA pump at 24 and 48 h after surgery for both the Group

Acu and Sham was normally distributed. There was no difference in pain intensity on VAS-10 cm between the two groups at 8 h, 24 h and 48 h postoperatively at rest and on movement (Figure 3). However, Group Acu required 37% and 31% less fentanyl than Group Sham at 24 h and 48 h after surgery, respectively (Figure 4).

Side effects and rescue medication

Incidence of analgesia-induced side effects such as nausea, vomiting and dizziness were sig-

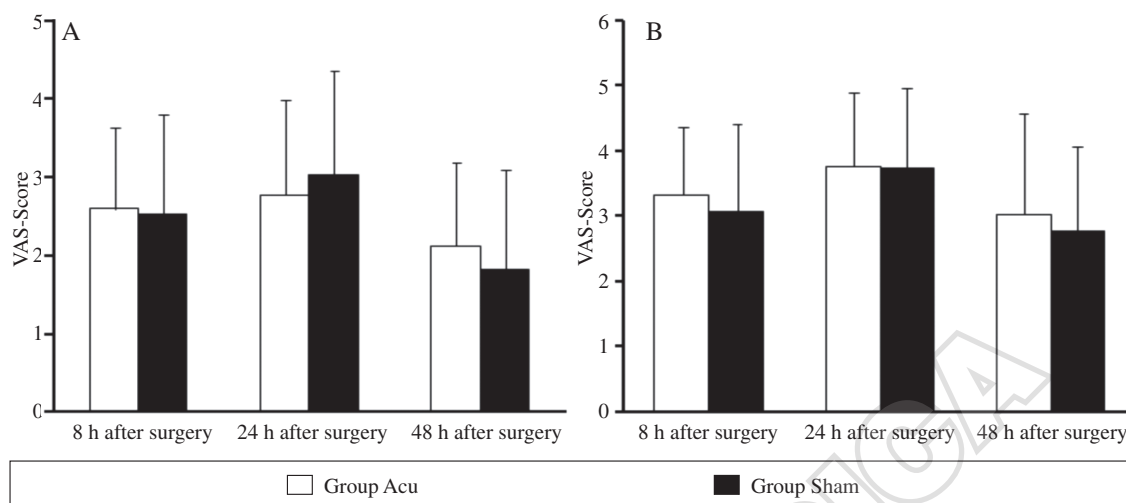


Figure 3.—Intensity of pain was measured at 8 h, 24 h and 48 h after surgery at rest and on movement. X- and Y-axis indicated different time points and VAS-10 cm scores postoperatively. A) There was no difference between two groups on pain intensity measured at 8 h, 24 h and 48 h after surgery at rest; B) there was no difference between two groups on pain intensity measured at 8 h, 24 h and 48 h on movement postoperatively. Group Acu: PCA+TENS on acupoints; Group Sham: PCA+Sham TENS on acupoints.

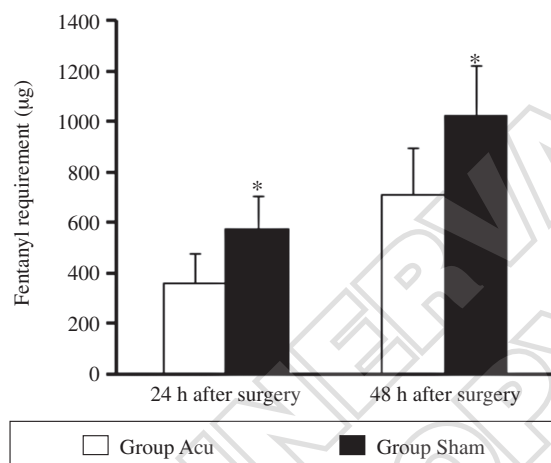


Figure 4.—Postoperative fentanyl requirement *via* PCA pump was measured at 24 h and 48 h after surgery. X- and Y-axis indicated different time points and fentanyl requirement postoperatively. Compared with Group Sham, fentanyl requirement in Group Acu was significantly lower at 24 h and 48 h after surgery. *= $P<0.001$ (using the unpaired Student's *t*-test). Group Acu: PCA+TENS on acupoints; Group Sham: PCA+Sham TENS on acupoints.

TABLE II.—Incidence of opioid related side effects and rescue medication.

Side effects	Group Acu	Group Sham
Nausea	5 (16.7)	14 (46.7) *
Vomiting	3 (10)	10 (32.3) *
Pruritus	3 (10)	4 (13.3)
Dizziness	2 (6.7)	8 (26.7) *
Excessive sedation	1 (3.3)	2 (6.7)
Rescue medication	4 (13.3)	11 (36.7) *

The incidence of opioid-related side effects and rescue medication for postoperative analgesia was significantly higher in Group Sham than in Group Acu. Values are numbers of patients (%); *= $P<0.05$ (using χ^2 test). Group Acu: PCA+TENS on acupoints; Group Sham: PCA+Sham TENS on acupoints.

nificantly higher in the Group Sham than the Group Acu. The incidence of pruritus and excessive sedation were similar in both groups. The frequency of rescue medication in group Acu was significantly lower than group Sham (Table II). There were no TENS-related side effects reported during the study.

Success of blinding

Twenty-seven patients in group Acu *vs.* 24 in the Group Sham affirmed that they had experienced true TENS on acupoints and intended to receive this method for pain relief in the future (Table III). Two patients from the Group Acu and three from the group Sham deemed that they received sham TENS. One patient from the Group Acu and three patients from the group Sham were not sure whether they had experienced the true TENS or sham. The differences in success of blinding between the two groups were not significant.

TABLE III.—*Assessment of blinding.*

Answer	Group A (N.=30)	Group Sham (N.=30)
TENS was true	27 (90)	24 (80)
TENS was sham	2 (6.7)	4 (13.3)
Unsure	1 (3.3)	2 (6.7)
Ready to re-experience TENS	27 (90)	24 (80)

There was no difference in both groups in assessment of blinding. Values are numbers of patients (%). Group Acu: PCA+TENS on acupoints; Group Sham: PCA+Sham TENS on acupoints.

Discussion

Undoubtedly, it is very important for elderly patients to receive effective postoperative pain management, as this can reduce the occurrence of postoperative complications, such as cardiopulmonary complications,^{3, 4} and postoperative cognitive dysfunction.⁶ However, the administration of systemic narcotic drugs for pain relief, commonly used in clinical practice, is not without side effects such as cardio-respiratory depression, nausea and vomiting. It has been proposed that the best postoperative pain management is not a single technique or drug but a balanced and multimodal analgesia.¹⁹

Postoperative pain after hip fracture is procedure-specific; THA is associated with lower pain compared with dynamic hip screws (DHSs) and intramedullary hip crews (IMHSs).²⁰ If non-pharmacological and non-invasive methods can be used as complements to the primary modality of pain relief, the complications associated with administration of opioid and non-opioid drugs, as well as the dosages, could be decreased. Previous studies have indicated that the use of complementary medicine for pain management after surgery including acupuncture and TENS is effective.^{21, 22} In line with our study, previous studies of TENS have also suggested that alternating low- and high-frequency, high intensity (9-20 mA) stimulation of acupoints is capable of decreasing postoperative pain and opioid consumption after surgery.^{10, 13}

It has been suggested that VAS scores should be less than 4 for satisfactory postoperative pain relief.²³ Although both groups patients experienced more pain at 24 h than at 48 h postoperatively which is likely to be due to frequency and

intensity of early rehabilitation after surgery,²⁰ the mean VAS scores in both groups reached to a satisfactory level in this study during the observation period up to 48 h after surgery. Hence, it is not surprising that in this study, the majority of patients in both groups were satisfied with postoperative analgesia but the patients who received TENS consumed much less fentanyl.

The precise analgesic mechanism of TENS remains largely unclear. However, according to the gate theory, TENS can suppress pain transmission mediated via sensory A- δ and C fibers.²⁴ An elegant study of electroacupuncture, which is quite similar to TENS on acupoints, showed that optimal frequencies can release met-enkephalin and dynorphins into the spinal fluid and cause synergistic pain relief with exogenous opioid medication.²⁵ Ma *et al.*²⁶ have reported that electroacupuncture reduces the levels of excitatory neurotransmitters responsible for pain modulation in the spinal cord in rats. It has also been reported that peripheral opioids were released after electroacupuncture in an inflammatory pain model in rats.²⁷ Interestingly, pre-incisional electropuncture has been shown to be an effective preemptive analgesic in a number of studies.^{28, 29} The proposed mechanism has been that preincisional stimulation yields specific effects on the production of pain-producing substances (*e.g.*, potassium and lactic acid) and the conductance of painful stimuli *via* κ -receptors by producing ir-dynorphin A.³⁰

It has been well documented that there is a dose-response relationship between opioid used and its related side effects.³¹ Therefore, any non-pharmacological methods that reduce the use of opioid medication are likely to be beneficial. Consistent with our observations, preoperative and postoperative TENS on specific acupoints have been reported to reduce opioid requirement for pain relief and its associated incidence of postoperative nausea and vomiting.³²

It should be pointed out that the major limitation of our study is not a truly double-blind study due to the nature of practicality.³³ This factor may distort the results accuracy by patients' unbiased judgment which is well documented in the literature.³⁴ Nevertheless despite this limitation, our result is in line with the other studies reported previously.^{35, 36}

Conclusions

Use of TENS on six specific acupoints in elderly patients undergoing total hip arthroplasty is a non-invasive, non-pharmacological, and simple technique for use in postoperative analgesia and is associated with reductions in opioid-related side effects. However, the relatively small sample size is a major limitation of this study. Further studies with larger sample size together with other types of surgery are needed to fully evaluate the efficacy of TENS for postoperative pain relief.

Key messages

— Transcutaneous electrical nerve stimulation (TENS) has been considered as a complementary intervention to relief various types of pain.

— This randomized controlled trial aimed to investigate the effect of TENS on acupoints for postoperative analgesia following total hip arthroplasty (THA) in elderly patients.

— Our data indicate that TENS can reduce postoperative analgesic requirement in elderly and hence decrease analgesic opioid related side effects.

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